

WILMS

Women's Interest-Free Loan Management System

Developer Guide — v1.8.0

WILMS Developer Guide

Version: 1.7.3 Audience: Engineers contributing to WILMS Classification: Confidential

1. Prerequisites

- Node.js 22.x
- npm (workspaces)
- Neon PostgreSQL account (or local DATABASE_URL)
- Git

2. Repository setup

```
git clone <repository-url> cd WILMS npm ci
```

Create apps/frontend/.env.local:

```
NEXT_PUBLIC_API_BASE_URL=/api/wilms NEXT_PUBLIC_USE MOCK=false  
DATABASE_URL=<neon-connection-string> WILMS_SESSION_SECRET=dev-only-change-me
```

Start development:

```
npm run dev
```

Application available at <http://localhost:3000>

3. Monorepo structure

| Package | Purpose |

|-----|-----|

| @wilms/frontend | Next.js UI + Route Handlers |

| @wilms/domain | Business logic, DB, HTTP app |

| @wilms/api | Thin adapter for dual-run |

| @wilms/shared-rbac | Roles and permissions |

| @wilms/shared-validation | Zod schemas |

| @wilms/shared-types | Shared TypeScript types |

4. Development modes

In-process (default)

Domain API runs inside Next.js Route Handlers. No separate server needed.

Dual-run (optional)

WILMS_API_MODE=proxy WILMS_API_UPSTREAM=http://127.0.0.1:4000 npm run dev:api # Terminal 1 — domain on :4000 npm run dev # Terminal 2 — frontend on :3000

4a. Location master

npm run db:apply:location-master -w @wilms/domain npm run db:apply:ghana-hierarchy -w @wilms/domain npm run seed:ghana-hierarchy -w @wilms/domain npm run db:backfill:locations -w @wilms/domain

Safe transactional reset (keeps users and RBAC):

WILMS_CONFIRM_DB_RESET=YES npm run db:reset:keep-users -w @wilms/domain

See documentation/location/.

5. Adding a domain module

1. Create module directory: packages/domain/src/modules/my-module/ 2. Add routes.ts with Express router 3. Add service.ts with business logic 4. Register router in packages/domain/src/http/app.ts 5. Add frontend service wrapper in apps/frontend/src/services/ 6. Add tests in packages/domain/src/tests/

Route pattern

```
export const myRouter = Router(); myRouter.use(requireAuth); myRouter.get('/my-resource',
requirePermission(PERMISSION.VIEW_REPORTS), asyncHandler(async (req, res) => {   sendData(res, await
myService.list());   }));
```

6. Authentication in development

Demo users in packages/domain/src/seed/demo-users.ts:

| Email | Password | Role |

|-----|-----|-----|

| admin@wilms.demo | DemoAdmin1! | Super Admin |

7. Database

Migrations

npm run db:generate -w @wilms/domain # Generate from schema changes npm run db:migrate -w @wilms/domain
Apply migrations npm run verify:migrations -w @wilms/api

Schema

Drizzle schema in packages/domain/src/db/schema/. Money columns as integer pesewas.

8. Testing

```
npm run test           # Frontend unit tests (sharded) npm run test -w @wilms/domain # Domain unit tests npm
run test:e2e          # Playwright E2E npm run smoke:rbac          # RBAC smoke npm run verify:api-integrity #
Route registration
```

9. Linting and type checking

npm run lint npm run type-check npm run verify:version

10. Frontend conventions

- App Router pages in apps/frontend/src/app/
- Feature modules in apps/frontend/src/features/
- Shared UI in apps/frontend/src/components/ui/
- API client in apps/frontend/src/lib/apiClient.ts
- Role protection via middleware.ts and RoleGuard

11. Financial code rules

Do not modify without explicit approval:

- Pesewas integer money handling
- Pool ledger formulas
- Payment immutability rules
- Admin fee enforcement
- Reconciliation logic

See docs/FINANCIAL_MODEL.md.

12. Documentation

Official library: documentation/. Generate PDF/DOCX:

npm run docs:generate

Architecture hub: docs/. Update docs/architecture/progress-tracker.md after meaningful changes.

Market-readiness notes (v1.8.0):

- Guarantor eligibility: packages/domain/src/modules/borrowers/guarantor-eligibility.ts
- Display IDs: packages/shared-utils/src/display-ids.ts
- Borrower updates: packages/domain/src/modules/borrower-updates/
- Migration: 0044_v180_borrower_update_requests.sql

13. Deployment

Production deploys via Vercel on merge. Environment variables configured in Vercel dashboard. See [documentation/operations/OPERATIONS_MANUAL.md](#).

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